

Addis Ababa University

College of Natural and Computational Science

Department of Physics

COMPUTATIONAL PHYSICS: Quiz II

Due on December 19, 2025

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Problem 1

```

1  PROGRAM problem1a
2  IMPLICIT NONE
3  INTEGER , PARAMETER :: N=4
4  REAL :: x0(0:N), y0(0:N), x(0:2*N), fx(0:2*N), dx=1, step=0.5
5  REAL :: fx1(0:2*N), a0, a1
6  INTEGER :: i, k
7
8  x0 = (/0, 1, 2, 3, 4/)
9  y0 = (/5, 9, 18, 25, 51/)
10
11 CALL range(x0,x)
12 CALL Linear_fit(y0,fx)
13 CALL least_square(x0, y0, x, fx1)
14
15
16 OPEN(UNIT=10, FILE="problem1a.csv", ACCESS="SEQUENTIAL", POSITION="APPEND", STATUS="
REPLACE")
17 DO k=0, 2*N
18     WRITE(10,FMT="(5f8.3)" ) x(k),  fx(k), fx1(k)
19 END DO
20 CLOSE(UNIT=10)
21
22
23
24 PRINT*,x
25 WRITE(*, FMT="(f6.2)") fx
26
27
28 END PROGRAM problem1a
29
30
31 SUBROUTINE range(x0,x)
32 IMPLICIT NONE
33 INTEGER, PARAMETER :: N=4
34 REAL, INTENT(IN) :: x0(0:N)
35 REAL :: x(0:2*N), step=0.5, dx=1
36 INTEGER :: i
37 DO i=0, 2*N
38     IF (MOD(i,2) == 0) THEN
39         x(i) = x0(i/2)
40     ELSE
41         x(i) = x0((i-1)/2) + step
42     END IF
43 END DO
44 END SUBROUTINE range
45
46 SUBROUTINE Linear_fit(y0,fx)
47 IMPLICIT NONE
48 INTEGER, PARAMETER :: N=4
49 REAL, INTENT(IN) :: y0(0:N)
50 REAL :: fx(0:2*N), step=0.5, dx=1
51 INTEGER :: k
52 DO k=0, 2*N
53     IF (MOD(k,2) == 0) THEN
54         fx(k) = y0(k/2)
55     ELSE
56         fx(k) = y0((k-1)/2) + step/dx*(y0((k+1)/2) - y0((k-1)/2))
57     END IF
58 END DO

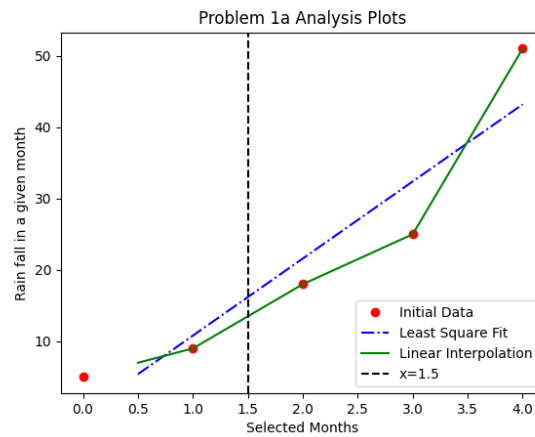
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59   fx(0) = y0(0)
60 END SUBROUTINE Linear_fit
61
62 SUBROUTINE least_square(x0,y0,x,fls)
63   IMPLICIT NONE
64   INTEGER, PARAMETER :: N=4
65   REAL, INTENT(IN) :: x0(0:N), y0(0:N), x(0:2*N)
66   REAL, INTENT(INOUT) :: fls(0:2*N)
67   REAL :: c1, c2, c3, c4, a0, a1
68   INTEGER :: i
69   c1 = SUM(x0)
70   c2 = DOT_PRODUCT(x0,x0)
71   c3 = SUM(y0)
72   c4 = DOT_PRODUCT(x0,y0)
73
74   a0 = (c1*c4 - c2*c3)/(c1**2 - (N+1)*c2)
75   a1 = (c1*c3 - (N+1)*c4)/(c1**2 - (N+1)*c2)
76
77
78   DO i=0, 2*N
79     fls(i) = a0 + a1*x(i)
80   END DO
81
82 END SUBROUTINE least_square
83
84

```

Listing 1: Fortran Code for Problem 1

Figure 1: Problem 1a solution; it shows plots of the initial data, least square fit, linear interpolation and the line $x = 1.5$

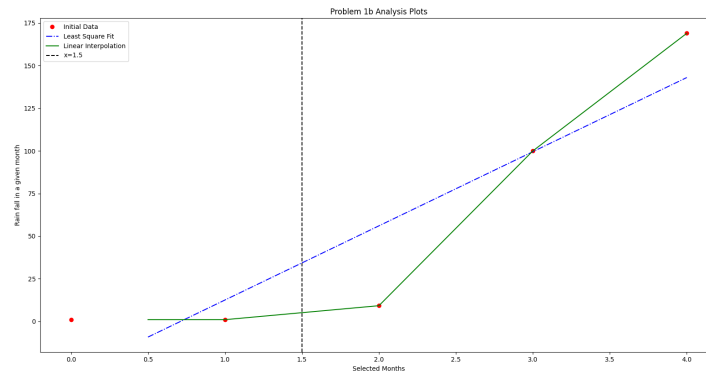


Figure 2: Problem 1b solution; it shows plots of the initial data, least square fit, linear interpolation and the line $x = 1.5$

Problem 2

```

1  FUNCTION f1(x) RESULT(y)
2      IMPLICIT NONE
3
4      REAL, INTENT(IN) :: x(:)
5      REAL :: y(size(x))
6
7      y = 1 + x + sin(x)
8
9  END FUNCTION
10
11
12
13 PROGRAM problem2a
14
15     IMPLICIT NONE
16     EXTERNAL :: f1
17     INTEGER, PARAMETER :: N=10000
18     REAL :: x(0:N), fx1(0:N), h, a=0, b=(3./4.)*3.14, S, Ss
19     INTEGER :: i, k, m
20     h = (b-a)/N
21     x(0) = a
22     DO i=1,N
23         x(i) = x(i-1) + h
24     END DO
25
26     fx1 = f1(x)
27     PRINT*, x
28     !PRINT*, fx1
29
30     S = 0
31
32     DO k=1, N
33         S = S + h/2*(fx1(k-1) + fx1(k))
34     END DO
35
36     PRINT*, S
37
38     Ss=0

```

```

39      DO k=1, N, 2
40          Ss = Ss + h/3*(fx1(k-1) + 4*fx1(k) + fx1(k+1))
41      END DO
42
43      PRINT*, Ss
44
45
46
47  END PROGRAM problem2a
48
49
50

```

Listing 2: Fortran Code for Problem 2

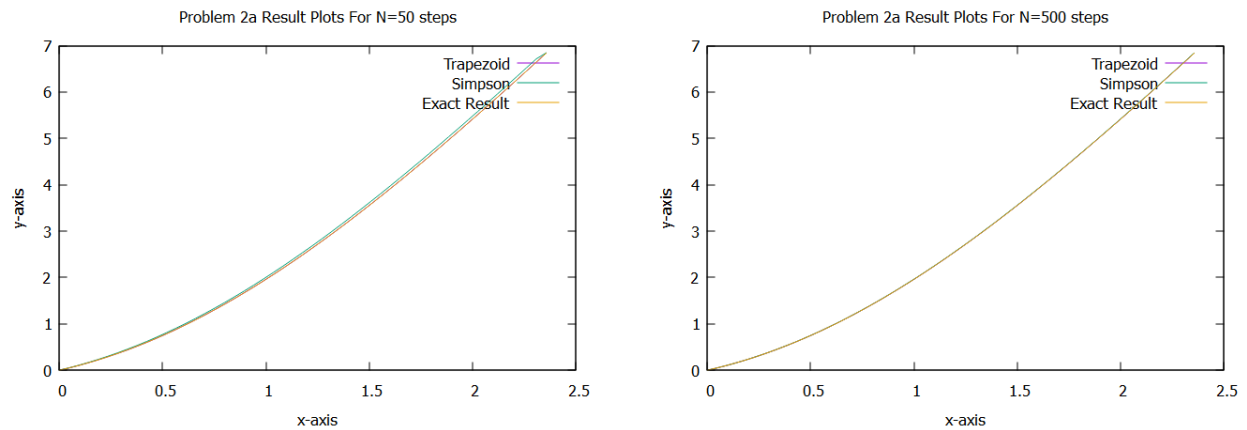


Figure 3: The results of problem 2a using Trapezoid and Simpson methods.

Problem 3

```

1  PROGRAM problem3a
2  IMPLICIT NONE
3  REAL, PARAMETER :: dx=0.001, a=0, b=(3./4.)*3.14
4  INTEGER, PARAMETER :: N=(b-a)/dx
5  REAL :: x(0:N), fx(0:N), fm(N), ff(N), fb(N)
6  INTEGER :: i, k
7
8  x(0) = 0
9  DO i=1, N
10     x(i) = x(i-1) + dx
11 END DO
12
13 fx = 1 + x + sin(x)
14
15 DO k=1, N
16     fb(k) = (fx(k) - fx(k-1))/dx
17 END DO
18
19
20 DO k=0, N-1
21     ff(k+1) = (fx(k+1) - fx(k))/dx
22 END DO

```

```

23
24     DO k=1, N-1
25         fm(k) = (fx(k+1) - fx(k-1))/(2*dx)
26     END DO
27
28     OPEN(UNIT=10, FILE="problem3a.csv", ACCESS="SEQUENTIAL", POSITION="APPEND", STATUS="
29     REPLACE")
30     DO k=0, N-1
31         WRITE(10,FMT="(4f14.8)") x(k), ff(k), fb(k), fm(k)
32     END DO
33     CLOSE(UNIT=10)
34
35     print*, ff(15), fm(15)
36
37 END PROGRAM problem3a
38

```

Listing 3: Fortran Code for Problem 3

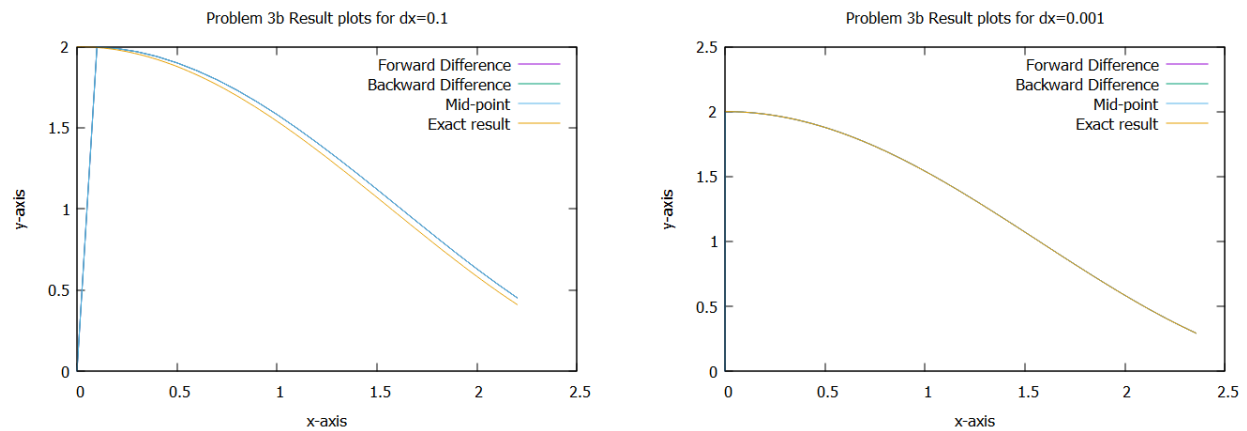


Figure 4: Differentiation result for problem 3a.

Problem 4

```

1  PROGRAM problem4
2  IMPLICIT NONE
3  INTEGER, PARAMETER :: N=10
4  REAL :: x0(1:N), y0(1:N), x(1:2*N), fx(1:2*N)
5  INTEGER :: i,k
6  x0 = (/ -0.83836, -0.383775, 0.0358423, 0.525396, 1.01495, 1.18979, 1.43457, 1.81921,
7  2.37870, 2.97316 /)
8  y0 = (/ 8.54662, 22.7661, 39.2307, 50.0075, 39.5300, 31.1480, 19.1738, 7.79823, 1.66143,
9  0.164646 /)
10
11 DO i=1, 2*N
12     IF (MOD(i,2).ne.0) THEN
13         x(i) = x0((i+1)/2)
14     ELSE
15         x(i) = (x0((i)/2) + x0((i+2)/2))/2
16     END IF
17 END DO
18
19 DO k=1, 2*N
20     IF (MOD(k,2).ne.0) THEN
21         fx(k) = y0((k+1)/2)
22     ELSE
23         fx(k) = fx((k-1)) + (x(k) - x((k-1)))/(x0((k+2)/2) - x0((k)/2)) * (y0((k+2)/2) - fx(
24 k-1))
25     END IF
26 END DO
27
28 OPEN(UNIT=10, FILE="problem4.csv", ACCESS="SEQUENTIAL", POSITION="APPEND", STATUS="
29 REPLACE")
30 DO k=1, 2*N - 1
31     WRITE(10,FMT="(2f8.4)") x(k), fx(k)
32 END DO
33
34 OPEN(UNIT=10, FILE="problem4r.csv", ACCESS="SEQUENTIAL", POSITION="APPEND", STATUS="
35 REPLACE")
36 DO k=1, N
37     WRITE(10,FMT="(2f8.4)") x0(k), y0(k)
38 END DO
39 CLOSE(UNIT=10)
40
41 PRINT*, fx
42 END PROGRAM problem4

```

Listing 4: Fortran Code for Problem 4

The data for problem 4 exhibit a gaussian peak around $x = 0.5$ with electron scattering intensity of near 50. There also a tail around for increasing the depth of atomic layer.

Problem 5

```

1  PROGRAM problem5
2  IMPLICIT NONE
3  INTEGER, PARAMETER :: N=5
4  REAL :: x0(1:N), y0(1:N), x(1:2*N), fx(1:2*N)
5  INTEGER :: i,k
6  x0 = (/ 0.17, 0.24, 0.38, 0.46, 0.89 /)

```

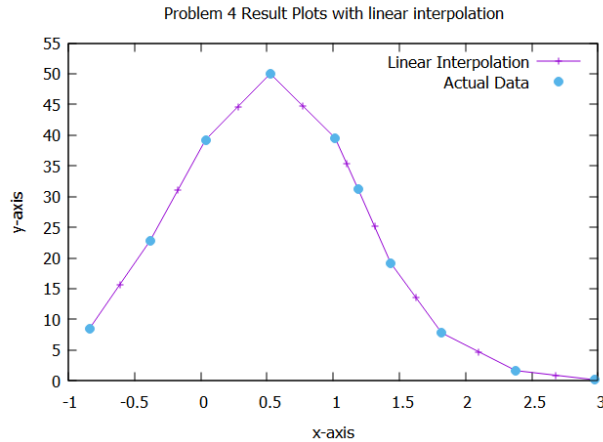


Figure 5: The linear fit and the actual data for problem 4.

```

7  y0 = (/500, 1000, 1200, 1600, 1800/)
8  DO i=1, 2*N
9      IF(MOD(i,2).ne.0) THEN
10         x(i) = x0((i+1)/2)
11     ELSE
12         x(i) = (x0((i)/2) + x0((i+2)/2))/2
13     END IF
14 END DO

15
16 DO k=1, 2*N
17     IF(MOD(k,2).ne.0) THEN
18         fx(k) = y0((k+1)/2)
19     ELSE
20         fx(k) = fx((k-1)) + (x(k) - x((k-1)))/(x0((k+2)/2) - x0((k)/2))*(y0((k+2)/2)-fx(
k-1))
21     END IF
22 END DO

23
24 OPEN(UNIT=10, FILE="problem5.csv", ACCESS="SEQUENTIAL", POSITION="APPEND", STATUS="
REPLACE")
25 DO k=1, 2*N - 1
26     WRITE(10,FMT="(2f14.5)") x(k), fx(k)
27 END DO

28
29 OPEN(UNIT=10, FILE="problem5r.csv", ACCESS="SEQUENTIAL", POSITION="APPEND", STATUS="
REPLACE")
30 DO k=1, N
31     WRITE(10,FMT="(2f14.5)") x0(k), y0(k)
32 END DO
33 CLOSE(UNIT=10)

34
35 PRINT*, fx
36
37 END PROGRAM problem5
38

```

Listing 5: Fortran Code for Problem 5

The data from problem 5 looks to be increasing for increasing humidity. However the rainfall seemed to approach some definite value for further increase in humidity. The estimated value at $x=0.69$ is found to be $y = 1706.97$

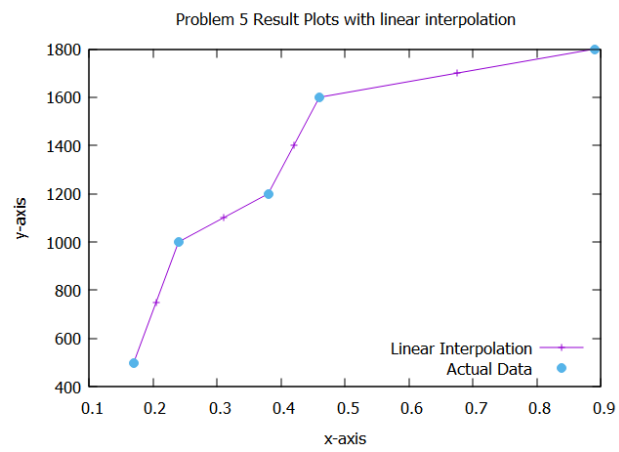


Figure 6: The linear fit and the actual data for problem 5.